

Overview of GIS Technology & Spatial Data Analytics

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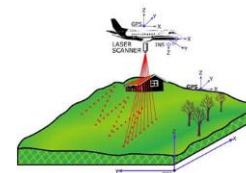
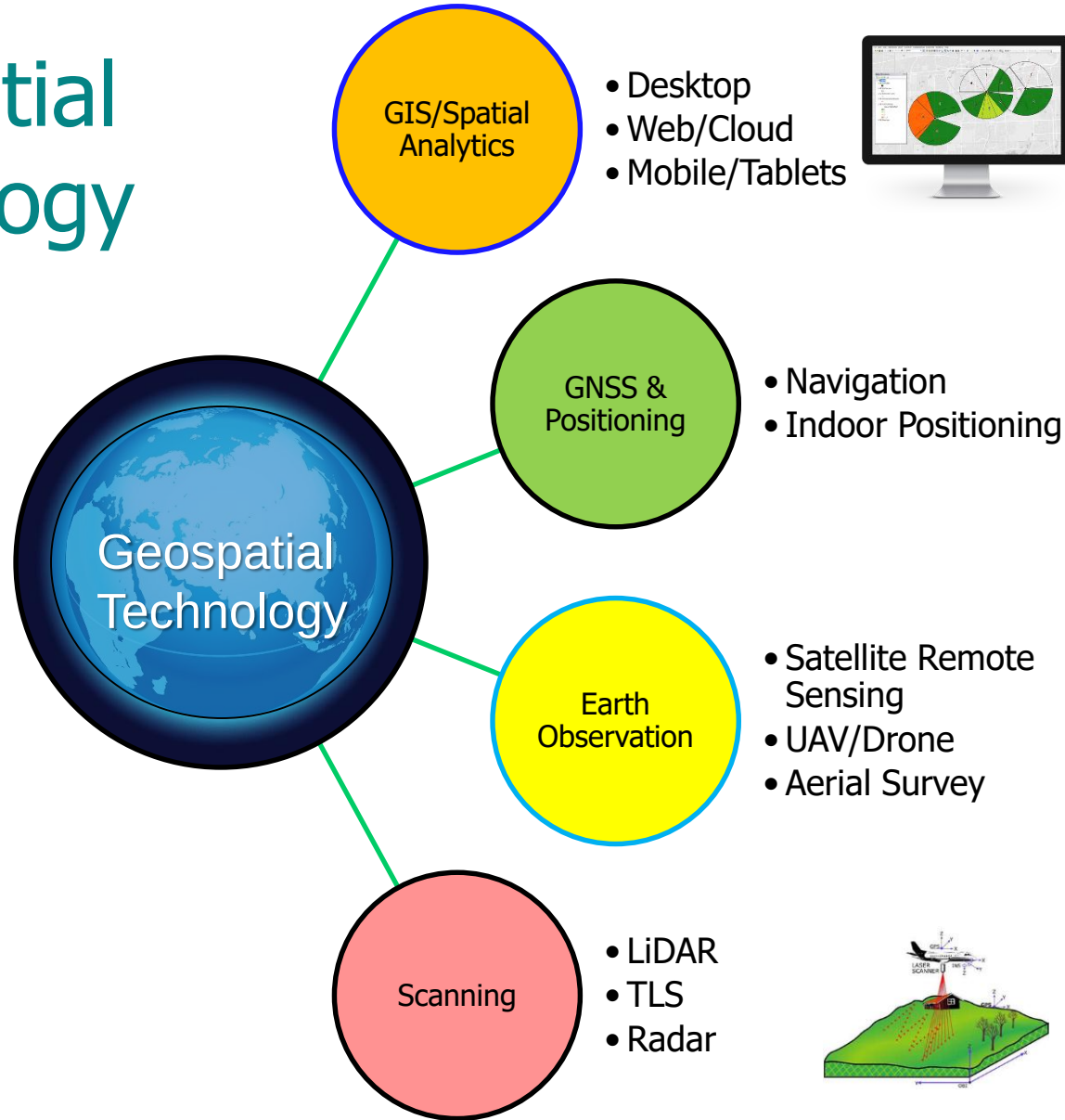
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Agenda of the Session

- Overview of Geospatial Technology
- Spatial Data Models
- Data Collection methods
- Data Cleaning
- Data Analytics
- Exploratory Data Analysis
- Concepts of Geo-processing
- Data Visualization tools
- Overview of AI/ML for spatial data analysis

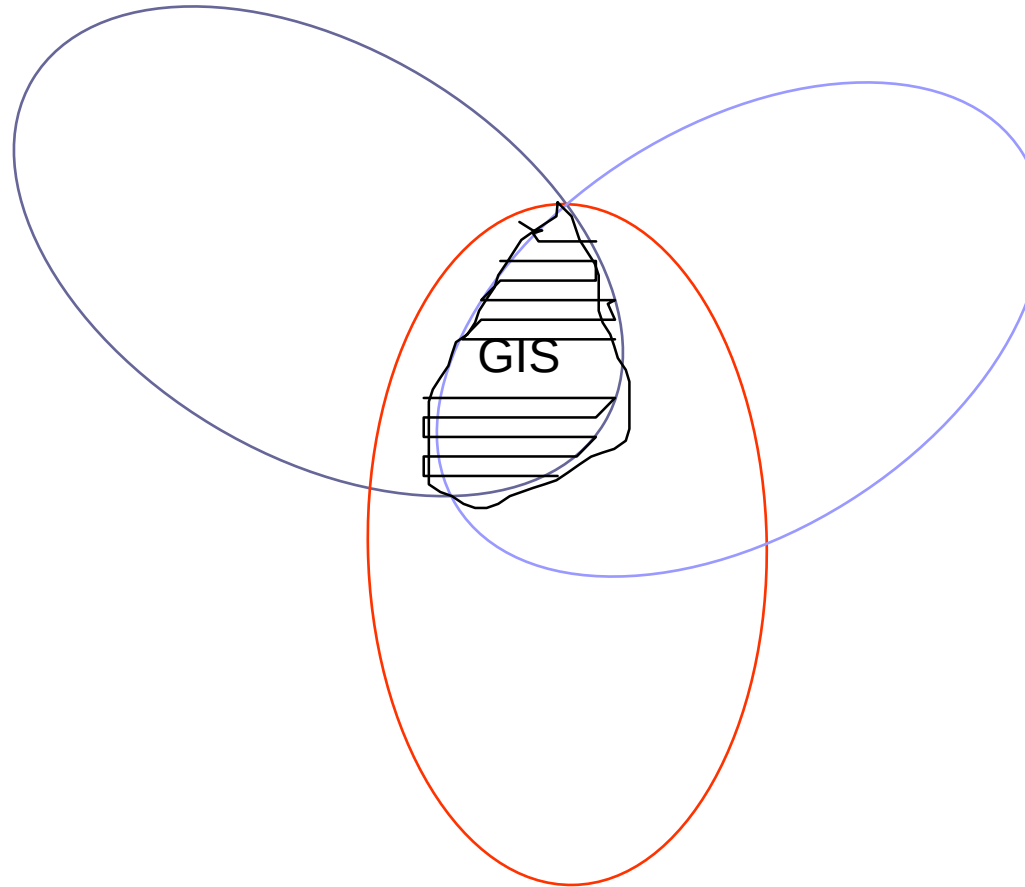
Geospatial Technology



Knowledge Base for GIS

Computer Science/MIS
 graphics
 visualization
 database
 system
 administration
 security

Geography and related:
 cartography
 geodesy
 photogrammetry
 landforms
 spatial statistics.



Application Area:
 public admin.
 planning
 geology
 mineral exploration
 forestry
 site selection
 marketing
 civil engineering
 criminal justice
 surveying

The convergence of technological fields and traditional disciplines.

Types of Data



Structured Data

Often numbers or labels, stored in a structured framework of columns and rows relating to pre-set parameters.

 ID CODES IN DATABASES

 NUMERICAL DATA GOOGLE SHEETS

 STAR RATINGS



Semi-structured Data

Loosely organized into categories using meta tags

 EMAILS BY INBOX, SENT, DRAFT

 TWEETS ORGANIZED BY HASHTAGS

 FOLDERS ORGANIZED BY TOPIC



Unstructured Data

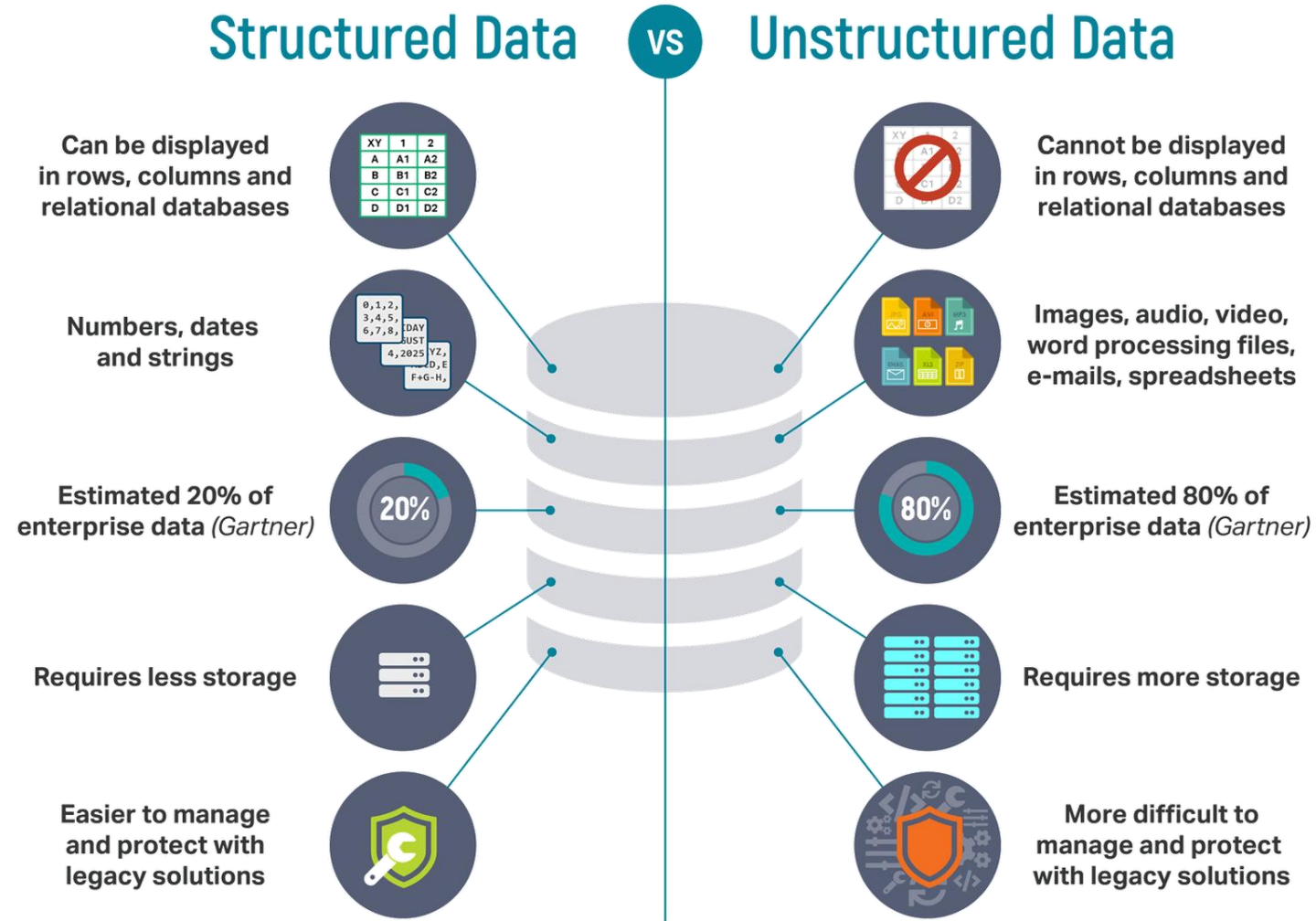
Text-heavy information that's not organized in a clearly defined framework or model.

 MEDIA POSTS, EMAILS, ONLINE REVIEWS

 VIDEOS, IMAGES

 SPEECH, SOUNDS

Understanding two important types data



What is Big Data?

- **Big data** is the term for a collection of unstructured data sets so large and complex that it becomes difficult to process using on-hand database management tools or traditional data processing applications.
- Challenges in handling Big data includes capture, curation, storage, search, sharing, transfer, analysis, and visualization.

Characteristics of Big Data



Representing Geographic Features

How do we describe geographical features?

- by recognizing two *types of data*:
 - **Spatial data** which describes location (where)
 - **Attribute data** which specifies characteristics at that location (what, how much, and when)

How do we represent these digitally in a GIS?

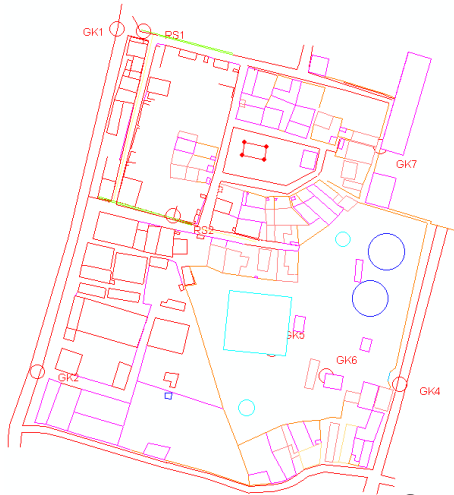
- by grouping into **layers** based on similar characteristics (e.g hydrography, elevation, water lines, sewer lines, grocery sales) and using either:
 - **vector** data model
 - **raster** data model
- by selecting appropriate **data properties** for each layer with respect to:
 - **projection, scale, accuracy, and resolution**

How do we incorporate into a computer application system?

- by using a relational Data Base Management System (DBMS)

Layers

Vector Layers



Line and Polygon feature



Buildings: polygons

Raster (image) Layer

Digital Ortho Photograph Layer:

Digital Ortho photo: combines the visual properties of a photograph with the positional accuracy of a map; in computer readable form.

Resolution: 0.5 meters

Accuracy: 1.0 meters

Scale:

1:500 Scale



Location: Gandhinagar Slum Hyderabad, Andhra Pradesh

Projection, Scale, Accuracy and Resolution

the key properties of spatial data

- **Projection:** the method by which the curved 3-D surface of the earth is represented by X,Y coordinates on a 2-D flat map/screen
 - distortion is inevitable
- **Scale:** the ratio of distance on a map to the equivalent distance on the ground
 - in theory GIS is scale independent but in practice there is an implicit range of scales for data output in any project
- **Accuracy:** how well does the database info match the real world
 - *Positional:* how close are features to their real world location?
 - *Consistency:* do feature characteristics in database match those in real world
 - is a road in the database a road in the real world?
 - *Completeness:* are all real world instances of features present in the database?
 - Are all roads included.
- **Resolution:** the size of the smallest feature able to be recognized
 - for raster data, it is the *pixel* size

The tighter the specification, the higher the cost.

Spatial Relationships

- **Topological relationships:**
 - adjacent, inside, disjoint, etc
- **Direction relationships:**
 - Above, below, north_of, etc
- **Metric relationships:**
 - "distance < 100"

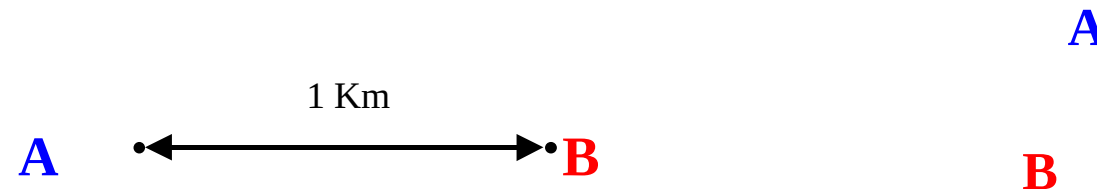
Operations to express the relationships

Spatial Relations

- **Topological Relations:** containment, overlapping, etc. [Egenhofer et al. 1991]



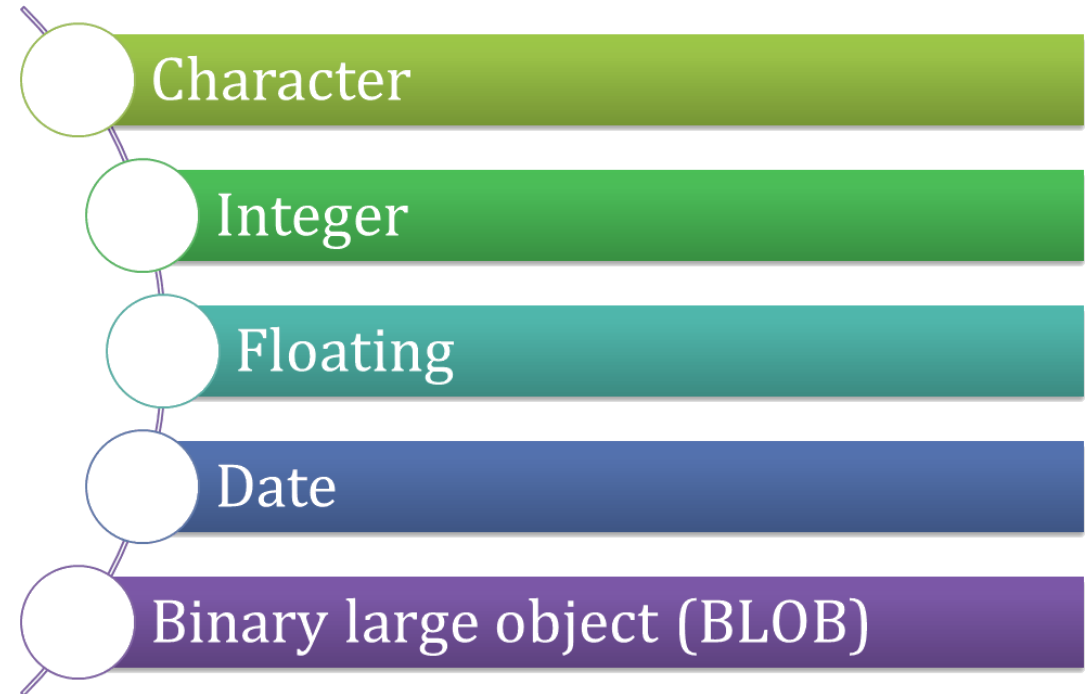
- **Metric Relations:** distance between objects, etc. [Gold and Roos 1994]



- **Direction Relations:** north of, south of, etc. [Hernandez et al. 1990; Frank et al. 1991]

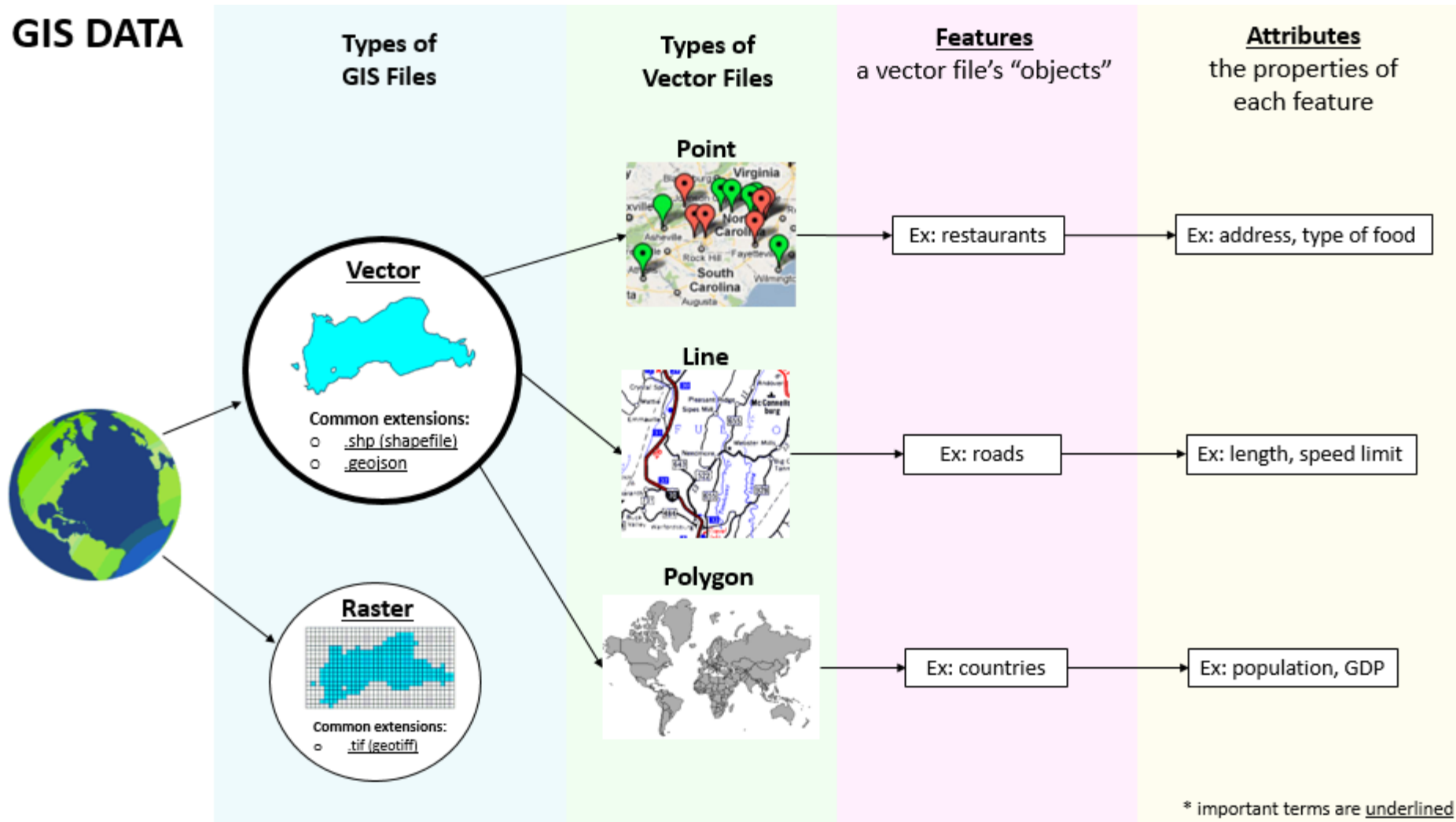
Attribute data types

- Represents the characteristics of Geographic features;
- Typically it is structured data which is stored in tabular format using standard data type;
- Database Management System (DBMS)/RDBMS are used for effective organization of attribute data.



GIS Data- Example

GIS DATA



GIS Data Analysis

Types of GIS Analysis

Type of Analysis	Description	Techniques	Examples
Spatial Analysis	Examines spatial relationships between features.	Buffering, Overlay, Spatial Join	Finding schools within 2 km of highways
Network Analysis	Studies movement, connectivity, and routes.	Shortest Path, Service Area.	Optimal delivery routes, emergency response planning
Surface Analysis	Analyzes continuous surfaces and terrain.	Slope, Aspect, Viewshed, Interpolation	Watershed analysis, solar potential mapping
Raster-Based Analysis (Arithmetic)	Uses grid/cell-based data for spatial computation.	Map Algebra, NDVI, Change Detection	Land cover change detection using satellite imagery
Vector-Based (Overlay) Analysis	Works with discrete features (points, lines, polygons).	Point Pattern, Polygon Overlay, Line Intersection	Mapping disease outbreaks, zoning compliance

Types of GIS Analysis

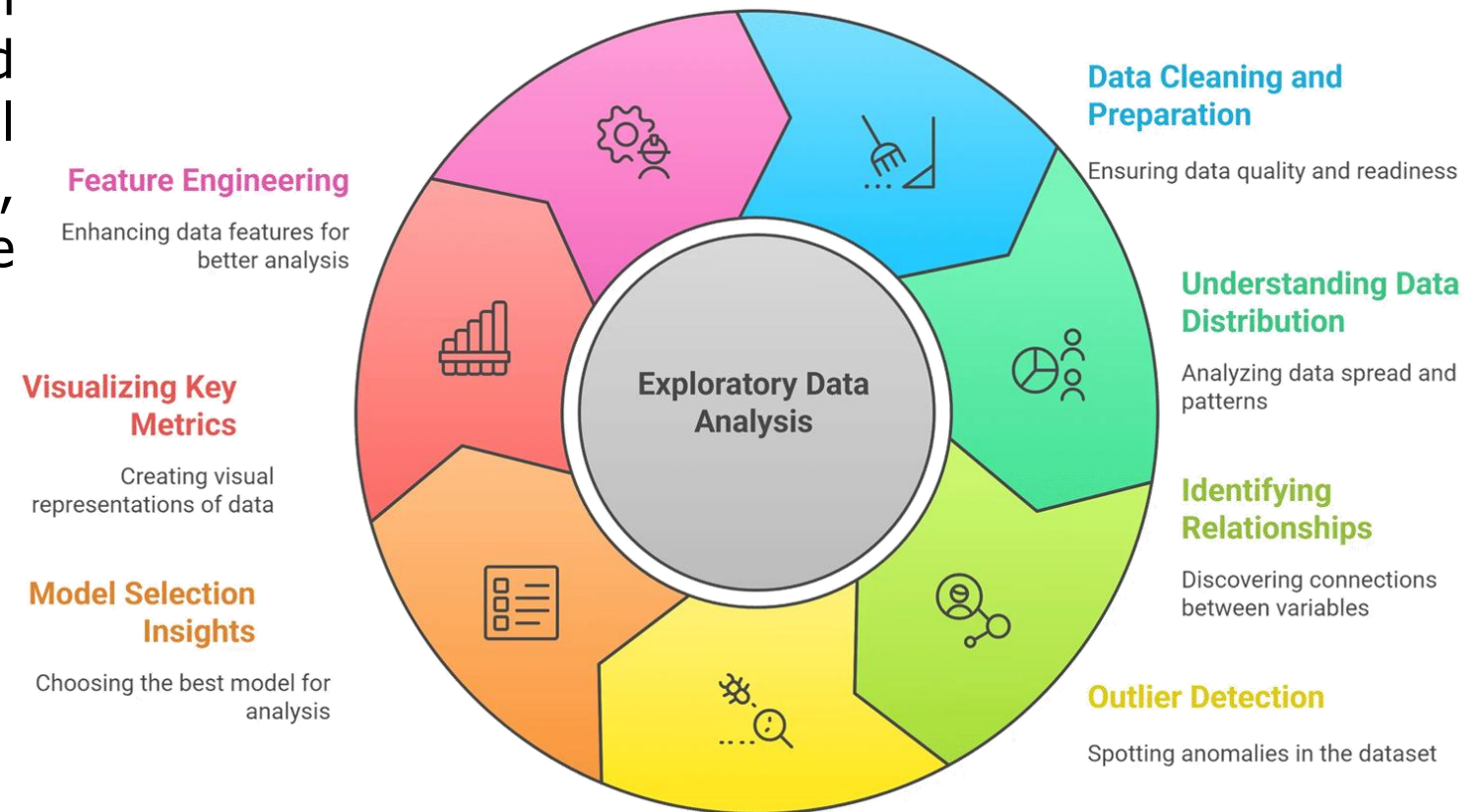
Type of Analysis	Description	Techniques	Examples
Geostatistical Analysis	Applies statistics to spatial data patterns.	Kriging, Hotspot Analysis	Predicting soil quality, crime hotspot mapping
Temporal Analysis	Studies changes over time in spatial datasets.	Time-Series Mapping, Trend Analysis	Urban sprawl monitoring, flood impact before & after
3D & Terrain Analysis	Visualizes and analyzes elevation or 3D data.	DEM Analysis, 3D Modeling, Fly-through	City 3D modeling, line-of-sight analysis
Predictive Modeling	Forecasts spatial phenomena using models.	Machine Learning, Simulation Models	Flood risk modeling, species habitat prediction
Image Analysis & Remote Sensing	Extracts information from imagery.	Image Classification, OBIA, Change Detection	Agricultural crop mapping, deforestation tracking

Exploratory Data Analysis (EDA) in GIS

Exploratory Data Analysis (EDA) in GIS is the process of **visually** and **statistically** examining geospatial datasets to uncover patterns, anomalies, and relationships before applying complex models:

Objectives:

- Understand spatial distribution & variability;
- Detect missing data & outliers;
- Identify clusters and spatial anomalies;
- Guide model selection and hypothesis building.



Techniques & Tools for EDA in GIS

Techniques:

- Descriptive statistics (mean, median, variance);
- Spatial visualization (choropleth, heatmaps, scatter plots);
- Histograms & density plots for attribute analysis;
- Spatial autocorrelation (Moran's I, Geary's C);
- Hotspot analysis for cluster detection.

Tools:

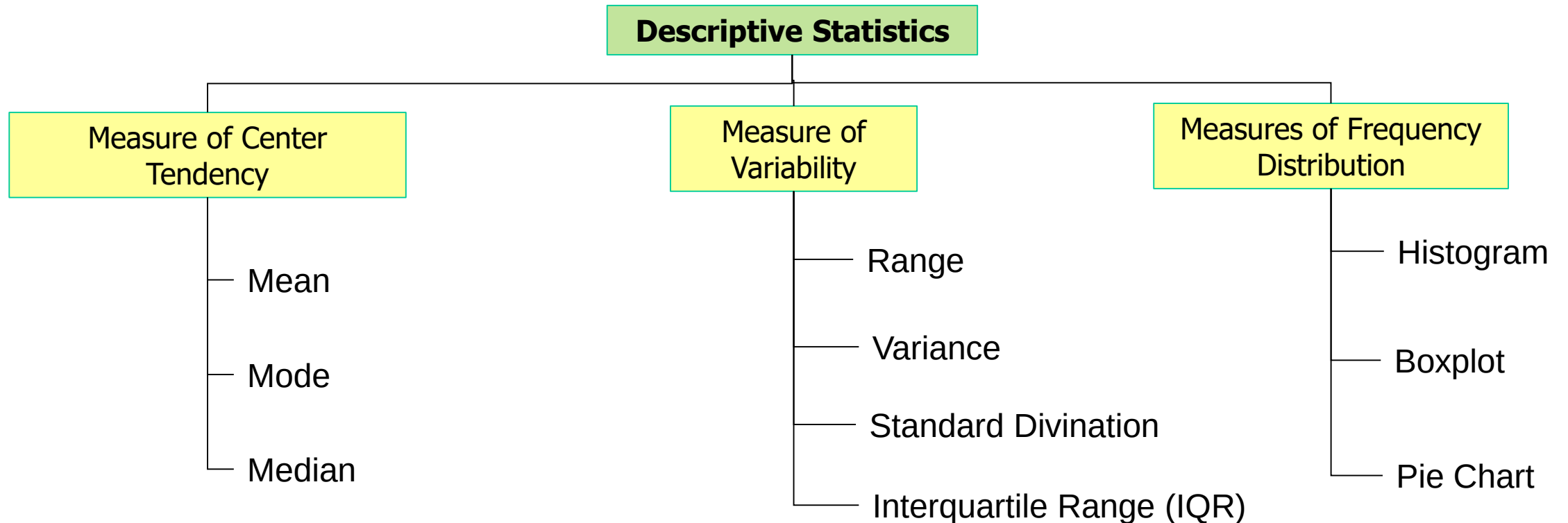
- **QGIS / ArcGIS Pro:** Spatial statistics, exploratory analysis toolbox
- **Python/R:** GeoPandas, PySAL, Folium, Matplotlib
- **Google Earth Engine:** Cloud-based EDA for large datasets

Outcome:

Insights for decision-making, improved spatial models, and better data quality assessment.

Descriptive statistics

Descriptive Statistics is a branch of statistics that summarizes and describes the main features of a dataset using **numerical measures, tables, and graphical methods**. In GIS, it is used to analyze **spatial data attributes** and provide a **quick understanding** of their **distribution** and **variability**.

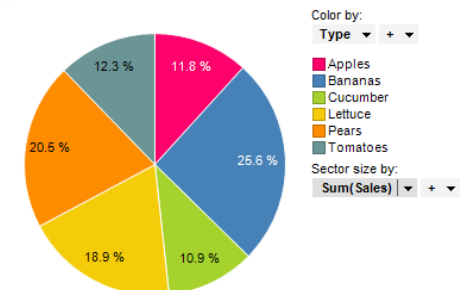
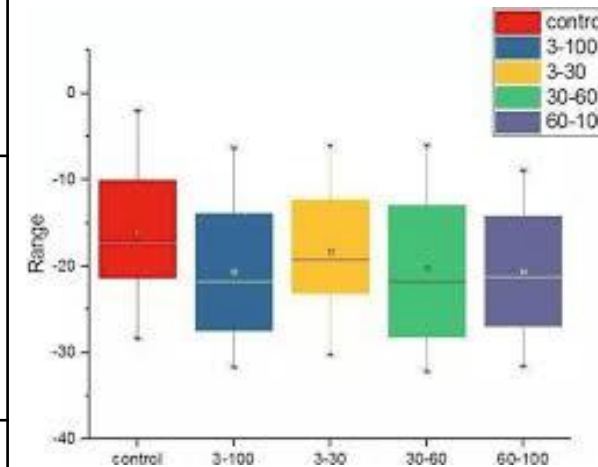
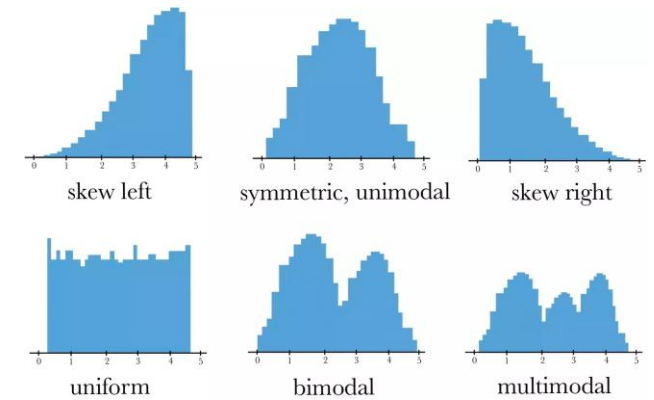


Descriptive statistics

Measure	Definition	Example
Mean (Average)	Sum of all values divided by total number of values.	Rainfall = [200, 220, 240] → Mean = 220 mm
Median	Middle value when data is arranged in order.	[180, 200, 220, 240, 260] → Median = 220 mm
Mode	Most frequently occurring value in a dataset.	[200, 220, 220, 240] → Mode = 220 mm
Range	Difference between maximum and minimum values.	Max = 500 m, Min = 200 m → Range = 300 m
Variance	Measure of how far data points spread from the mean.	Rainfall variance = 2025 mm ²
Standard Deviation (SD)	Square root of variance; indicates typical deviation from the mean.	Variance = 2025 → SD = 45 mm (average squared deviation)
Interquartile Range (IQR)	Spread of the middle 50% of data (Q3 – Q1).	Q1 = 190 mm, Q3 = 260 mm → IQR = 70 mm

Descriptive statistics

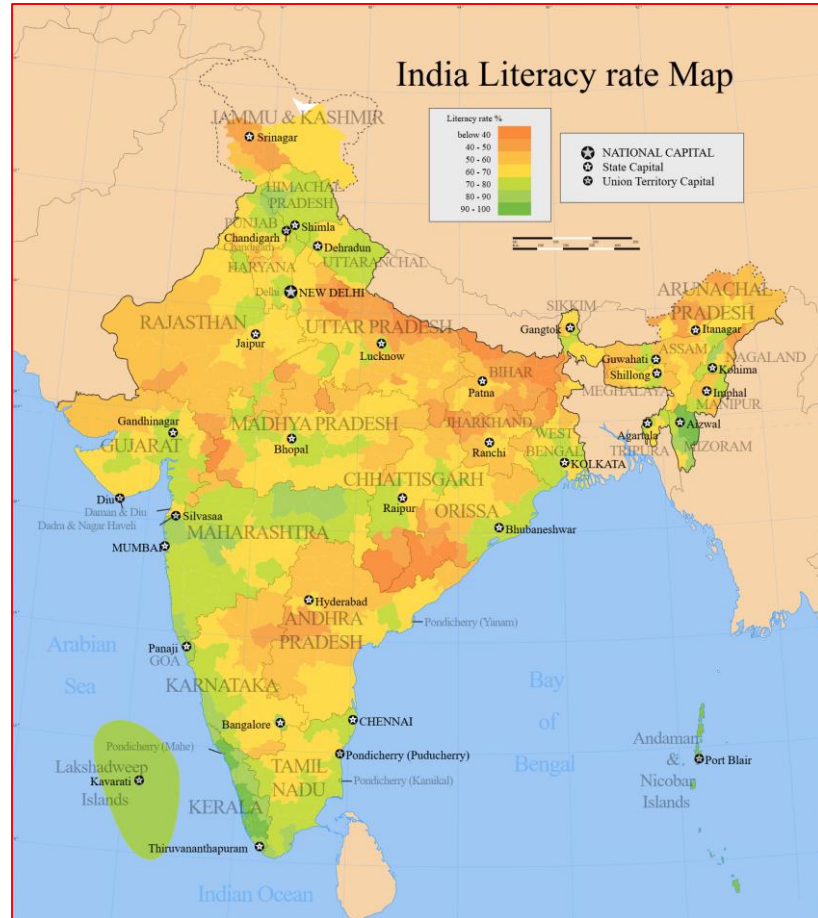
Measure	Definition (One Line)	Example
Histogram	Displays the frequency distribution of continuous data using adjacent bars.	Elevation values of a region grouped into 50 m bins.
Boxplot	Shows data distribution, median, quartiles, and outliers.	NDVI values for croplands with whiskers and outliers.
Pie Chart	Represents categorical data as proportional slices of a circle.	Land Use/Land Cover classes (Forest 40%, Urban 30%, Agriculture 20%, Water 10%).



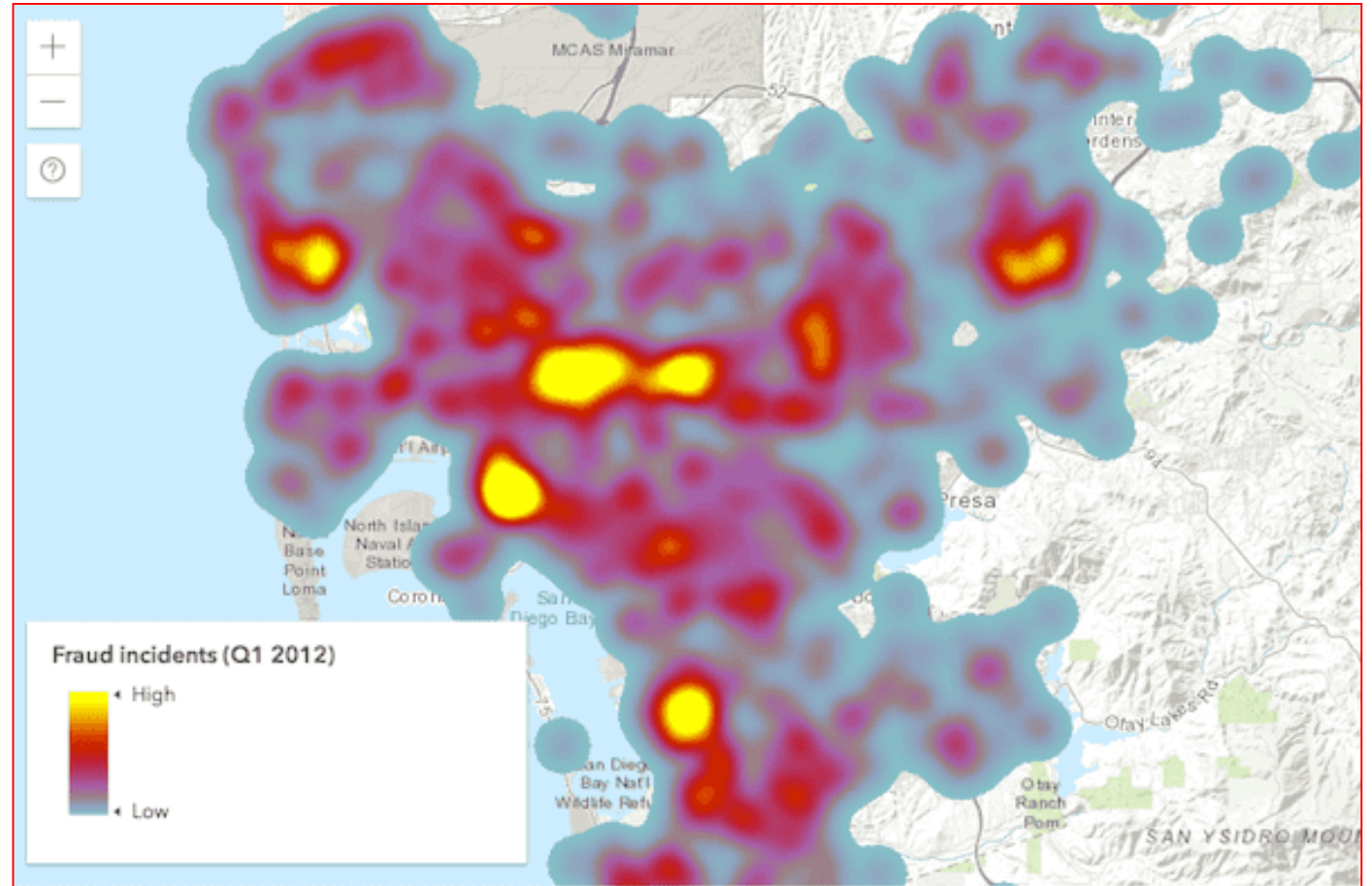
Spatial visualization

Technique	Definition	Example (GIS Context)
Choropleth Map	Uses colors or shades to represent attribute values.	Population density by district.
Heatmap	Visualizes intensity or density of points.	Crime hotspots in a city.
Dot Density Map	Represents quantities/events with dots.	Distribution of schools in a region.
Proportional Symbol Map	Uses scaled symbols to indicate magnitude of data.	City population sizes represented by circles.
3D Visualization	Displays data in three dimensions (X, Y, Z).	Urban 3D building height visualization.

Spatial Visualization

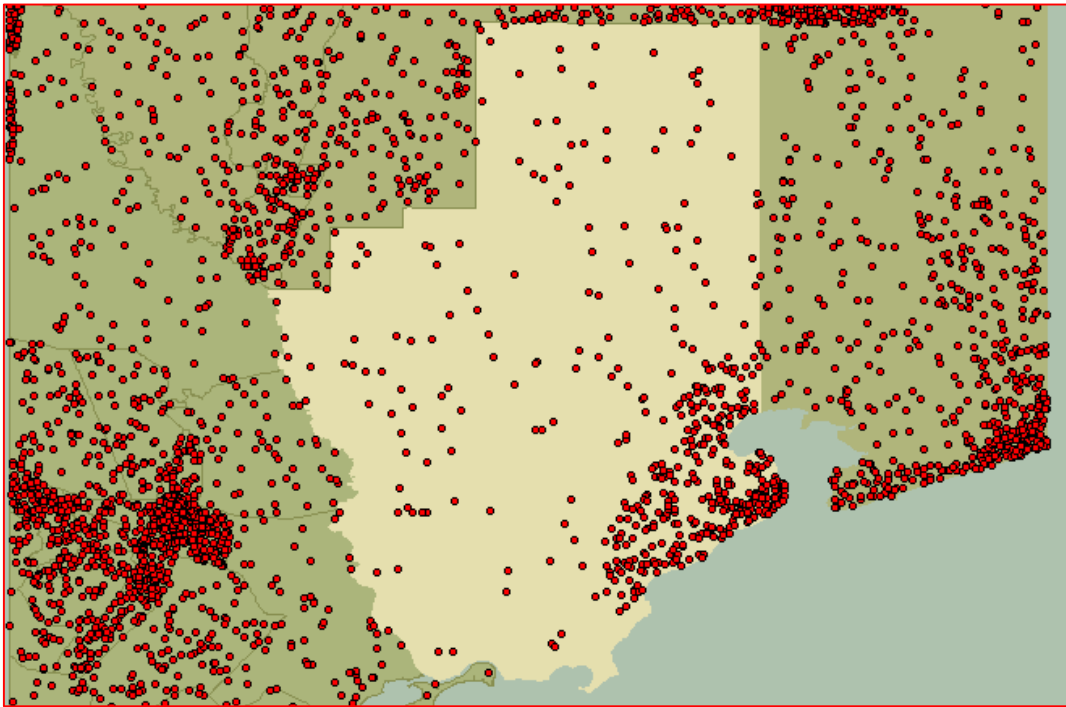


Choropleth Map

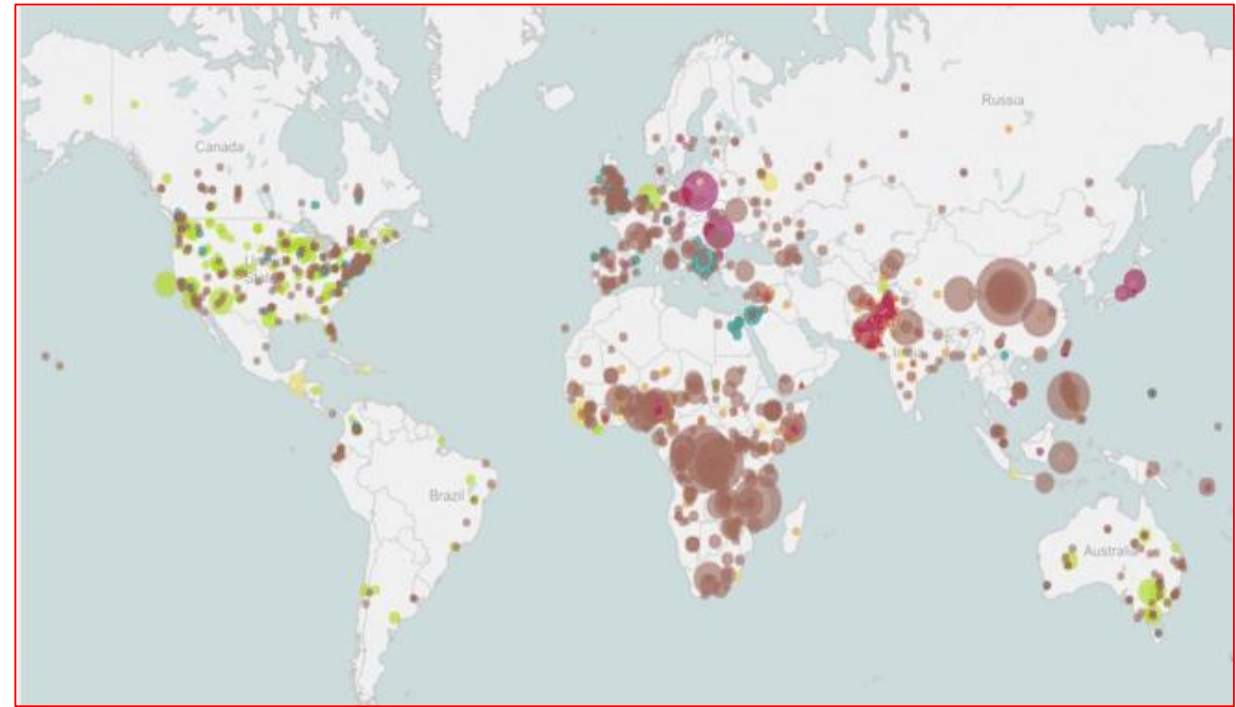


Heatmap

Spatial Visualization



Dot Density Map



Proportional Symbol Map

3D Visualization



Spatial autocorrelation

Spatial autocorrelation measures the **degree to which spatial features and their associated values are correlated in space**. It determines whether similar values occur near each other (clustered), are dispersed, or randomly distributed.

Positive Spatial Autocorrelation

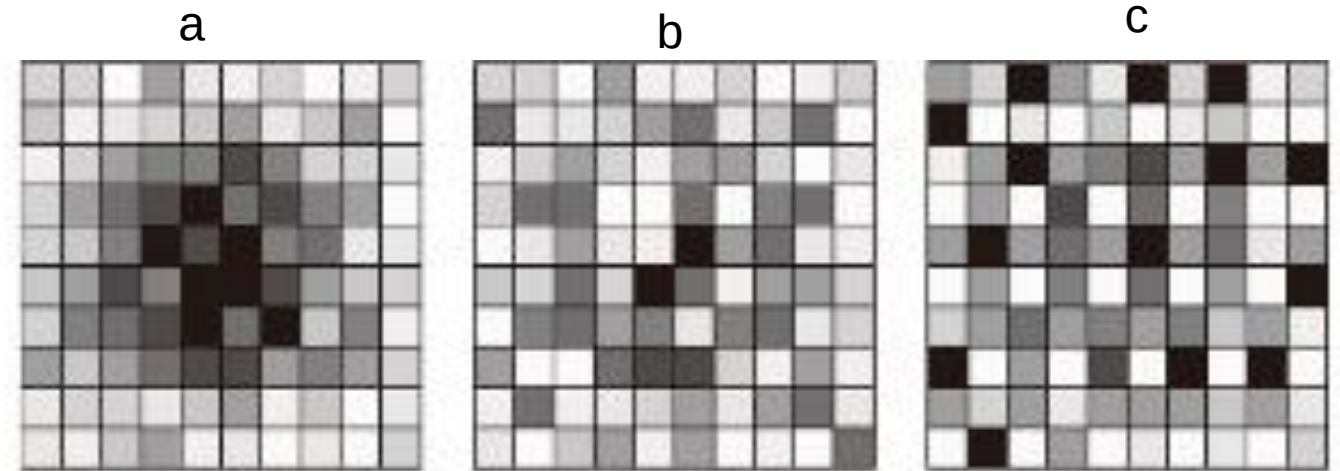
- Similar values cluster together.
- *Example:* High rainfall areas surrounded by other high rainfall areas.

Negative Spatial Autocorrelation

- Dissimilar values are neighbors.
- *Example:* High-income neighborhoods adjacent to low-income ones.

No Spatial Autocorrelation

- Random distribution with no pattern.



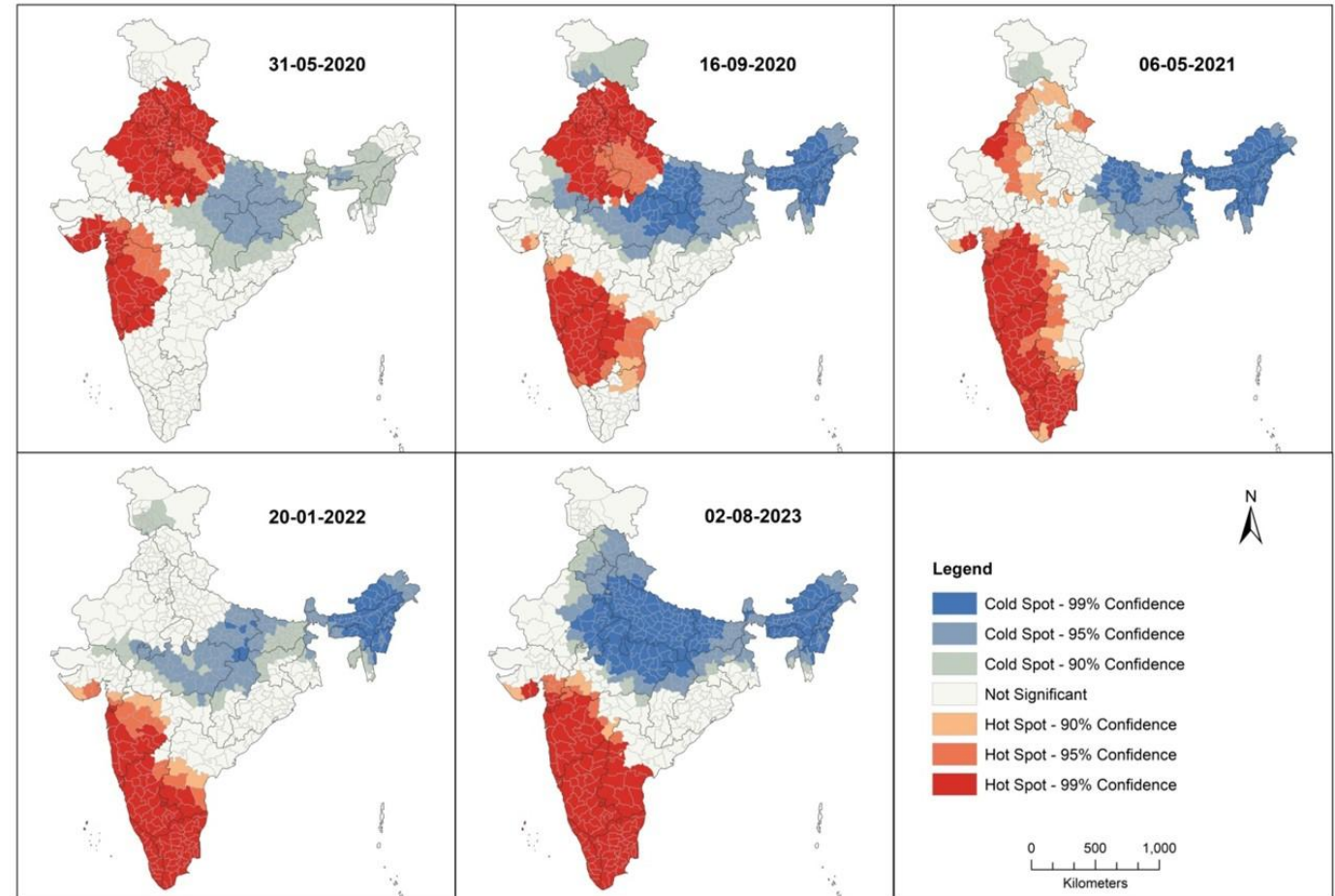
(a) Positive spatial autocorrelation. (b) Spatial randomness. (c) Negative spatial autocorrelation.

Hotspot analysis- Cluster Detection

Identifies statistically significant clusters of high (hot) or low (cold) values in spatial data.

Types of Clusters:

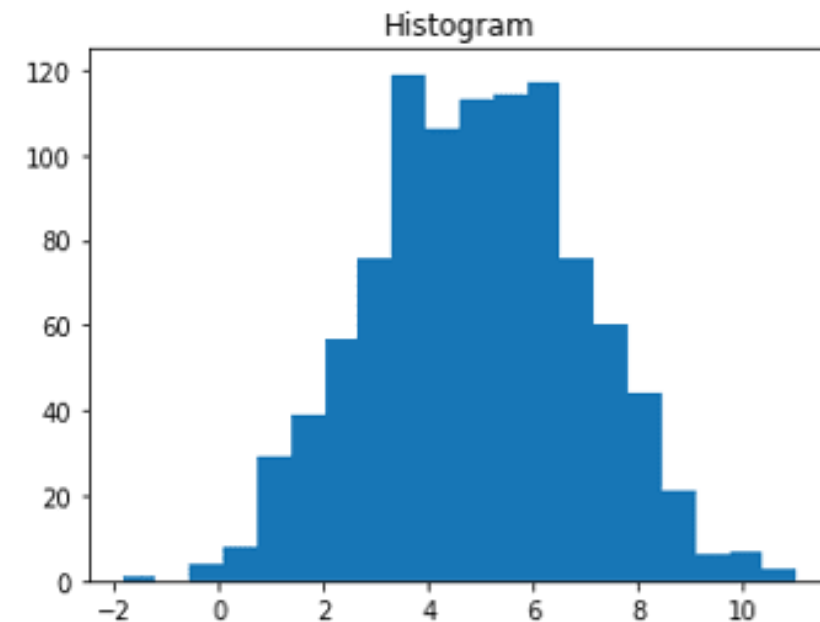
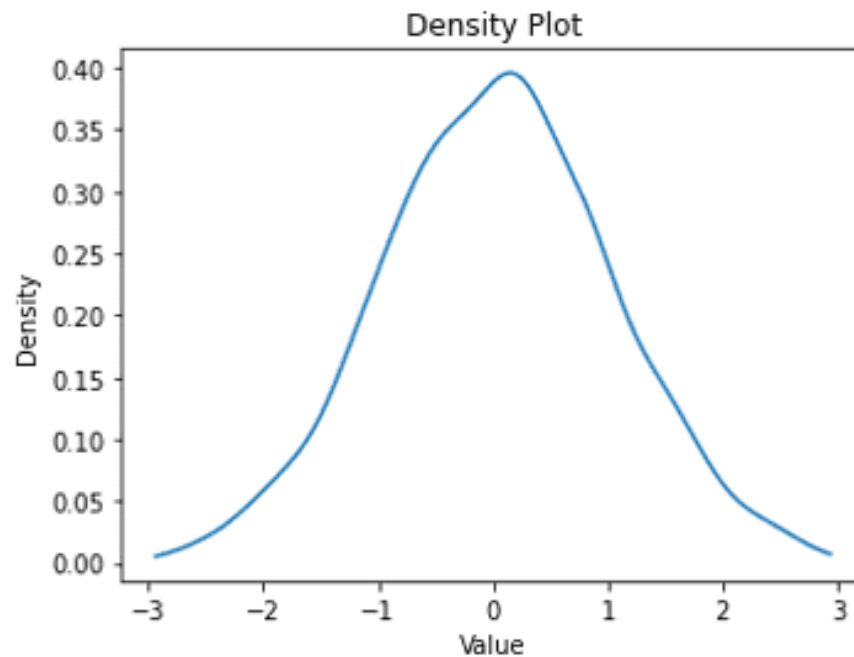
- **Hotspots:** High-value clusters
- **Coldspots:** Low-value clusters



Spatio-temporal Analysis of COVID-19 Hotspots in India

Histograms & density plots

Plot Type	Definition	Example (GIS Context)	Purpose
Histogram	Graph showing the frequency distribution of continuous data using adjacent bars (bins).	Elevation values divided into 50 m intervals.	Identify distribution shape, outliers, and concentration.
Density Plot	Smooth curve representing the probability distribution of a continuous variable (based on kernel density estimation).	NDVI values of cropland areas showing peaks of vegetation health.	Visualize data distribution trends more smoothly than histogram.



UNDERSTANDING GEOPROCESSING

What is Geoprocessing?

- Is a Computation technique used to compare, analyze, or manipulate map layers and their underlying geographic data to create new sets of data or information.
- Geoprocessing techniques for Raster and vector data analysis are used for various thematic application including natural resource management, disaster monitoring and management, governance applications etc.
- Typically geo-processing is used for automation of GIS analysis and information extraction methods.

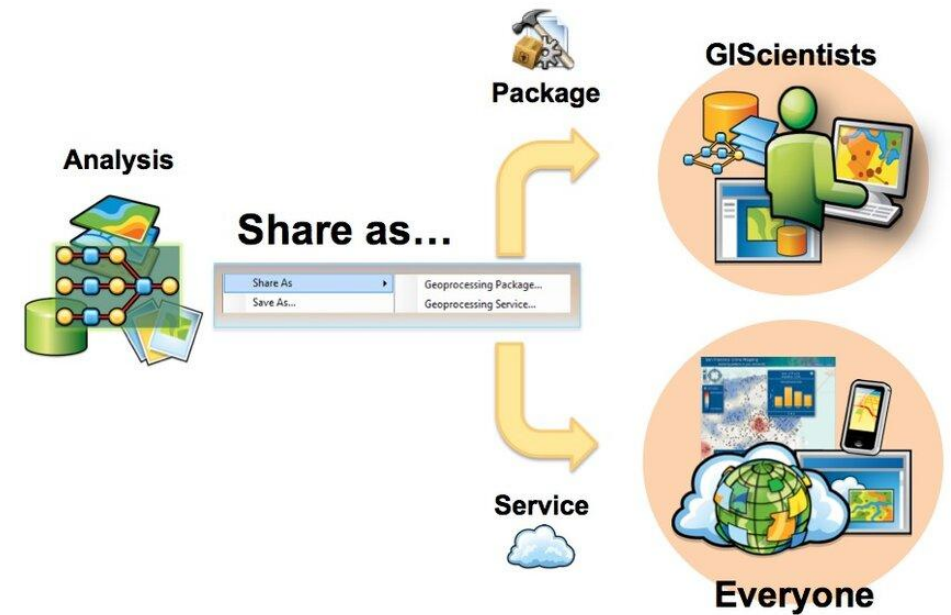
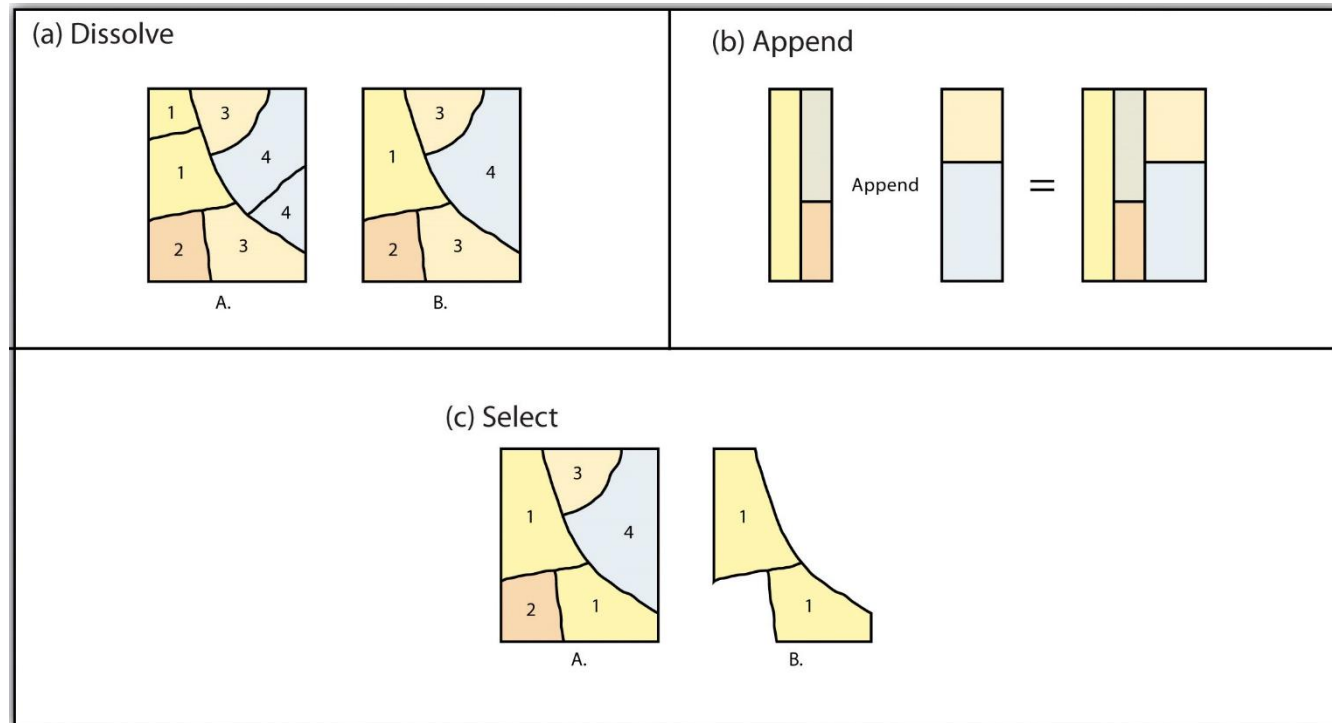
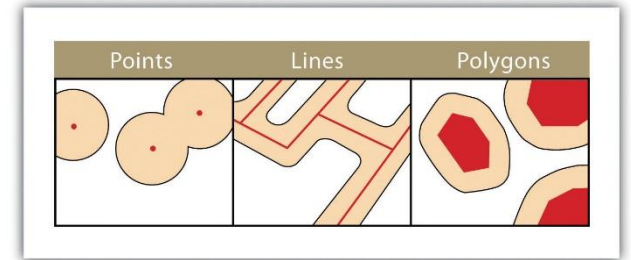


Image Source - ESRI

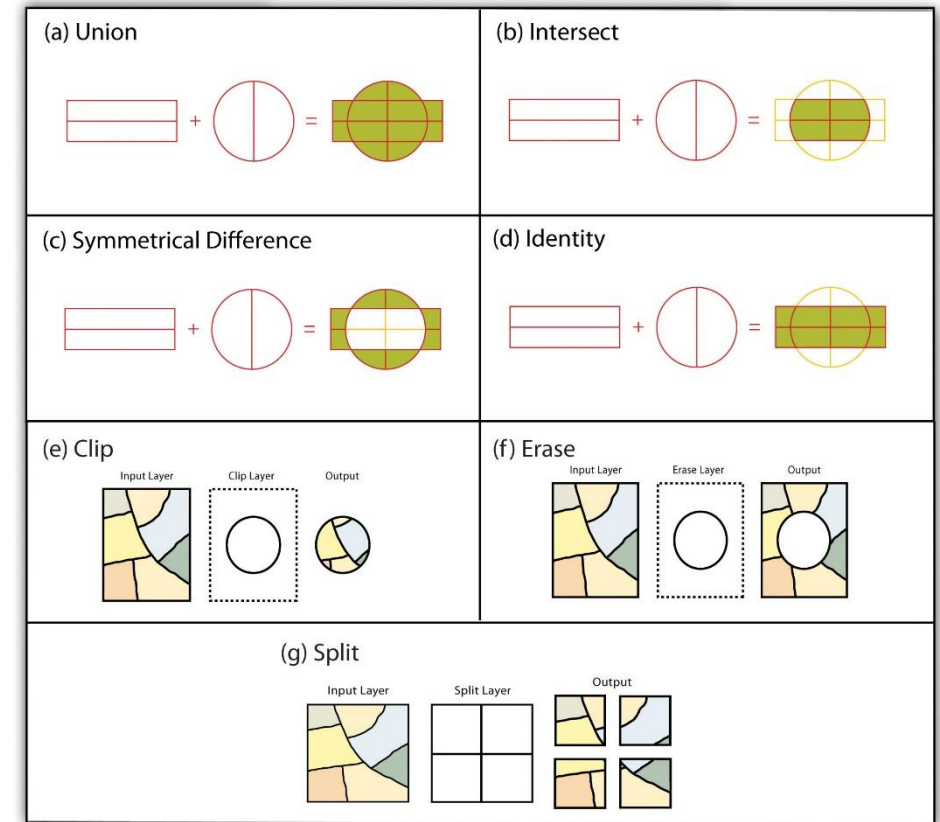
Geo-processing with Vector Data

- Single Layer Operations- Example- Buffering or proximity analysis;
- Single Layer Geo-processing:
 - dissolve operation ;
 - append operation ;
 - select operation ;
 - merge operation;



Geo-processing with Vector Data (Cont.)

- Multi Layer Operations- Typically Overlay analysis are performed;
- Application- Site suitability analysis
- Basic GIS overlay process:
 - Point-in-polygon; Polygon-on-point; Line-on-line; Line-in-polygon; Polygon-on-line; and Polygon-on-polygon.
- Basic GIS overlay operations: Boolean operators: AND, OR, and XOR:
 - **Union** overlay method employs the OR operator;
 - **Intersection** overlay method employs the AND operator;
 - **Symmetrical difference** overlay method employs the XOR operator;
 - **Identity** (also referred to as "minus") overlay method creates an output layer with the spatial extent of the input layer.

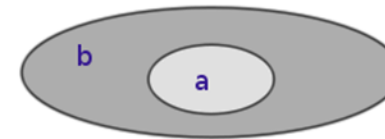


Geo-processing with Vector Data (Cont.)

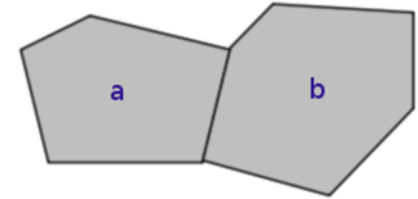
Spatial Join:

Join one dataset to another, attributes from the first dataset (join feature) are appended to the attributes in the second dataset (target feature) based on the relative spatial relationship between the two datasets i.e. geometries.

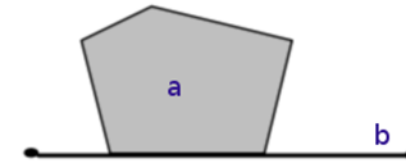
Within(a,b)



Touches(a,b)



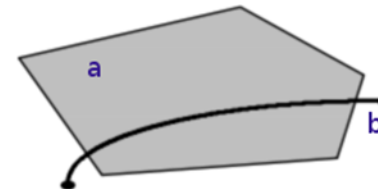
Touches(a,b)



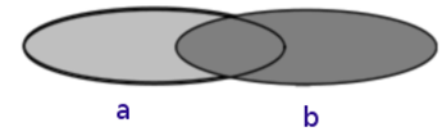
Crosses(a,b)



Crosses(a,b)



Overlaps(a,b)

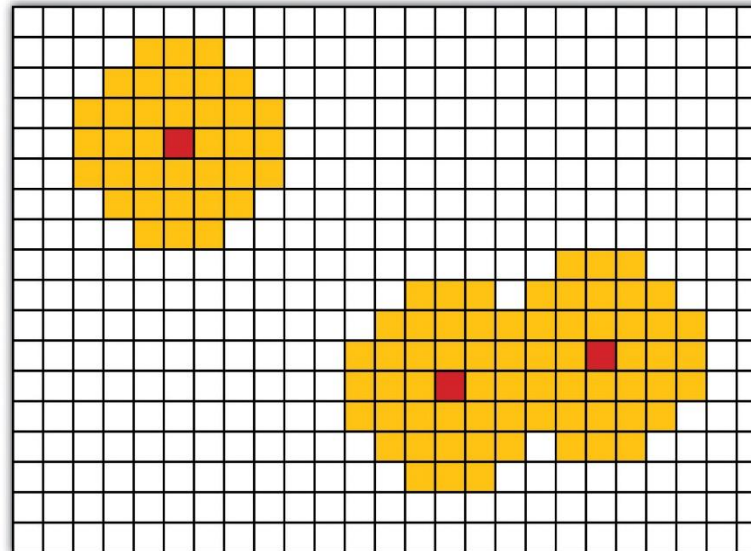


Geo-processing with Raster Data

- Raster data analysis enforces its spatial relationships solely on the location of the pixel. Raster operations performed on multiple input raster datasets generally output pixel values that are the result of computations on a pixel-by-pixel basis.
- Typically arithmetic operations are performed with raster data.

- **Single Layer Analysis**

- Reclassifying, or recoding;
- Raster Buffering.



Input Raster

456	416	364	326	243
448	364	315	276	218
359	325	268	234	164
306	296	201	133	44
274	231	184	65	5



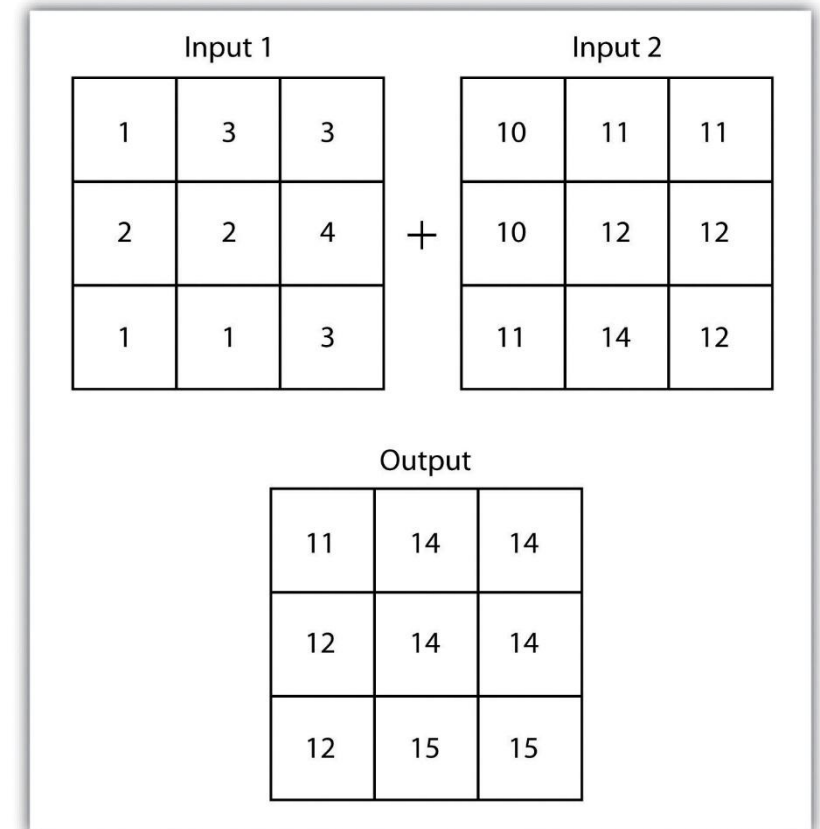
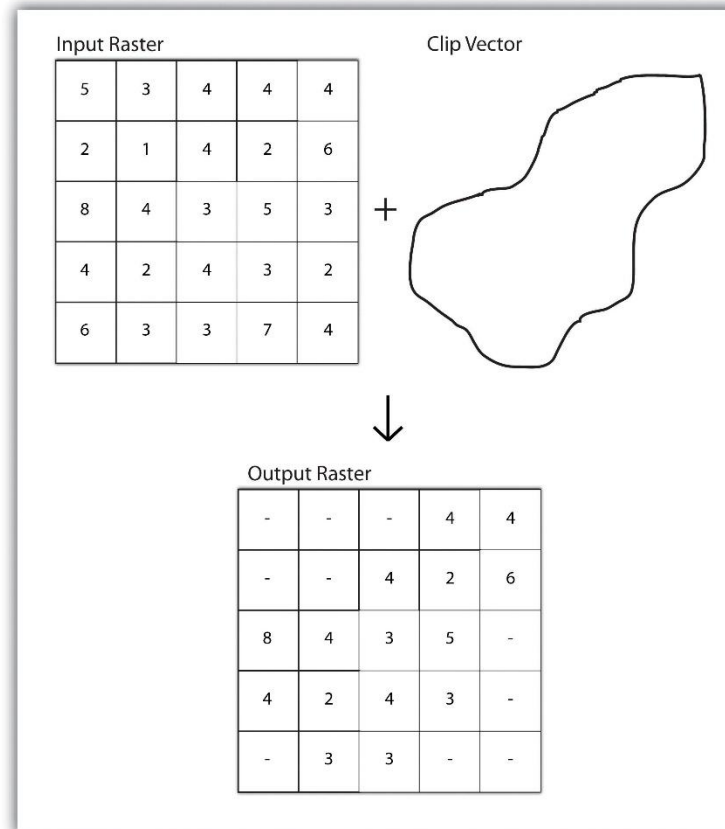
Reclassified Raster

5	5	4	4	3
5	4	4	3	3
4	4	3	3	2
4	3	3	2	1
3	3	2	1	1

Geo-processing with Raster Data (cont.)

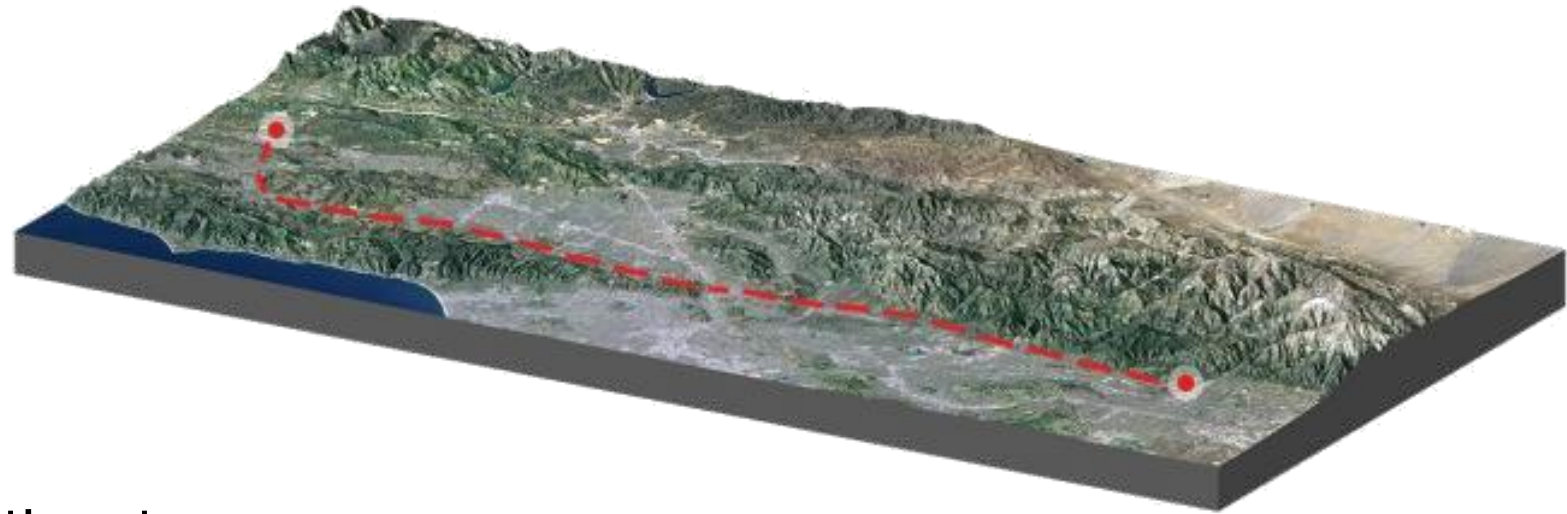
Multi-layer Layer Analysis

- Clip operation;
- Mathematical operations.



Raster analysis techniques

- Map Algebra;
- Zonal Statistics ;
- Contours ;
- Math Functions;
- Conditionals;
- Cost Path;
- Terrain Analysis;
- Suitability;
- Raster Processing Interpolation etc.



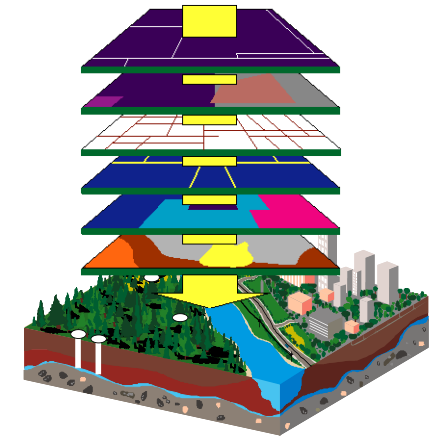
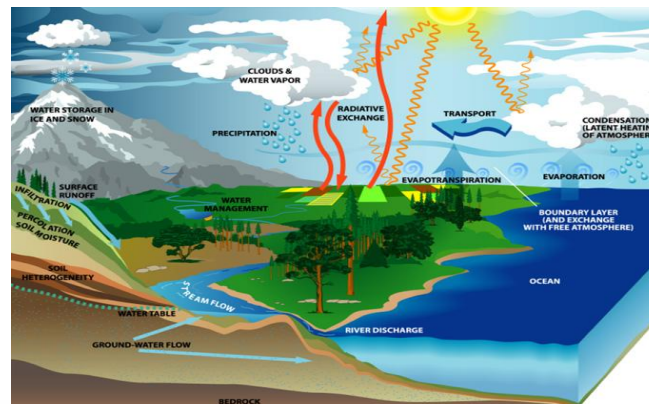
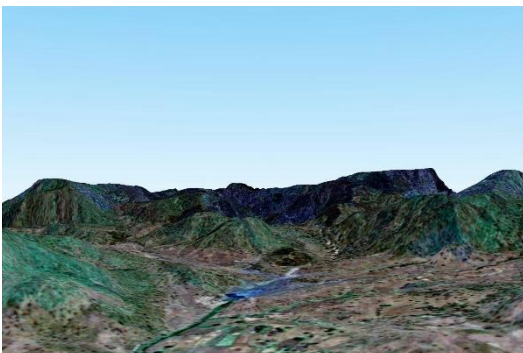
Geo-Modelling

What is Model?

“Abstract Representation of Reality.”

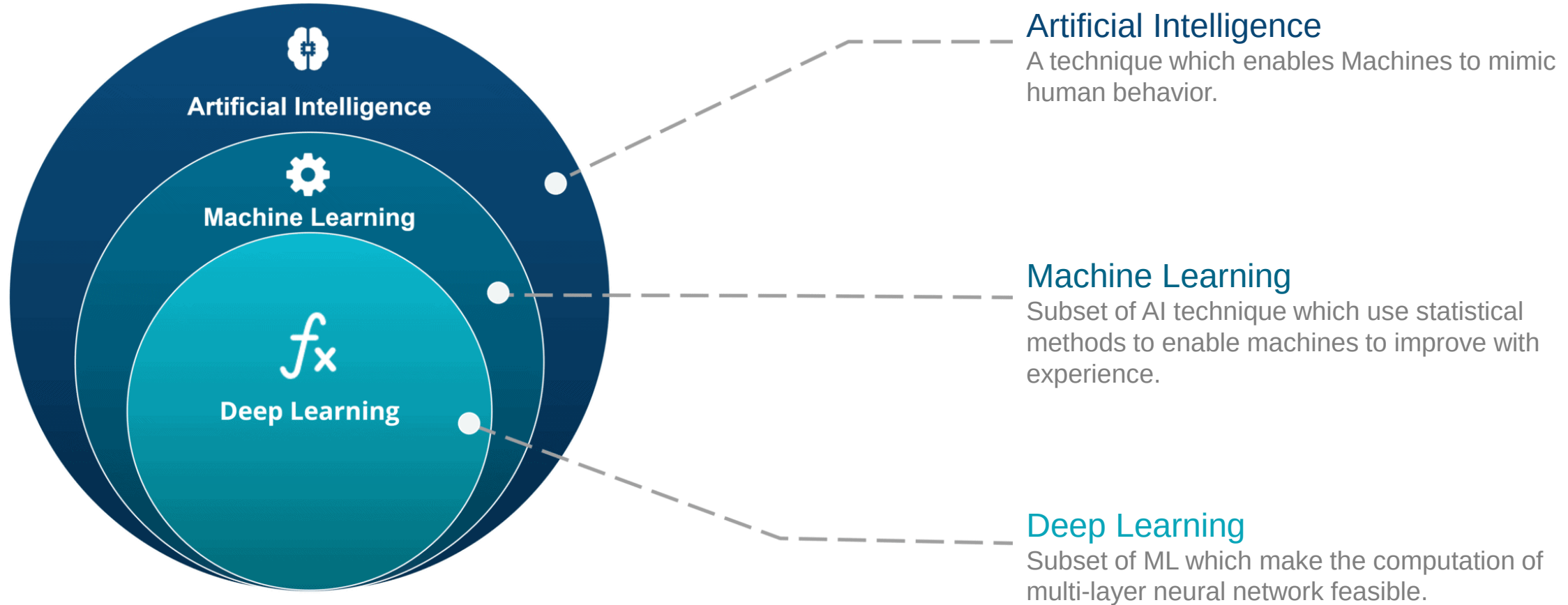
Spatial Model

- A model of some process operating in space (and time)
 - there is variation across the space (and through time)
 - location is important
 - the results of modeling change when locations change
 - locations must be known



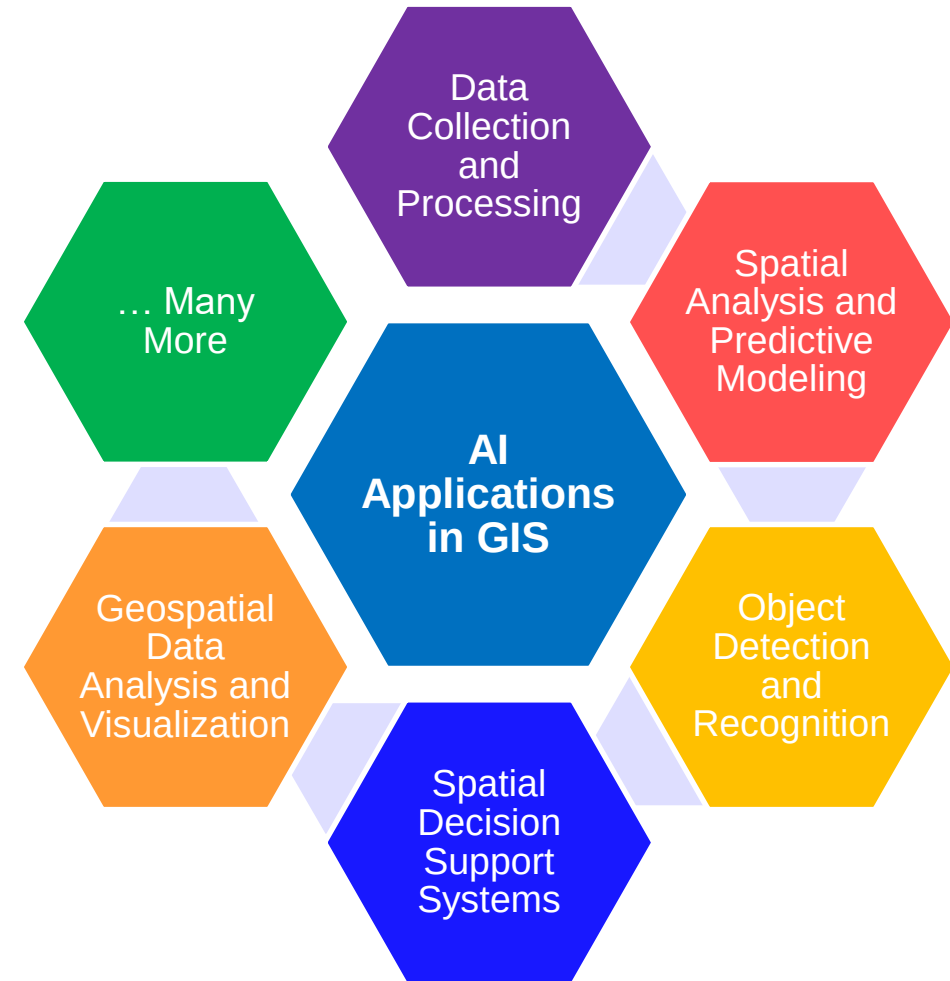
AI/ML/DL FOR GEODATA ANALYTICS

AI, ML and DL based algorithms are Important



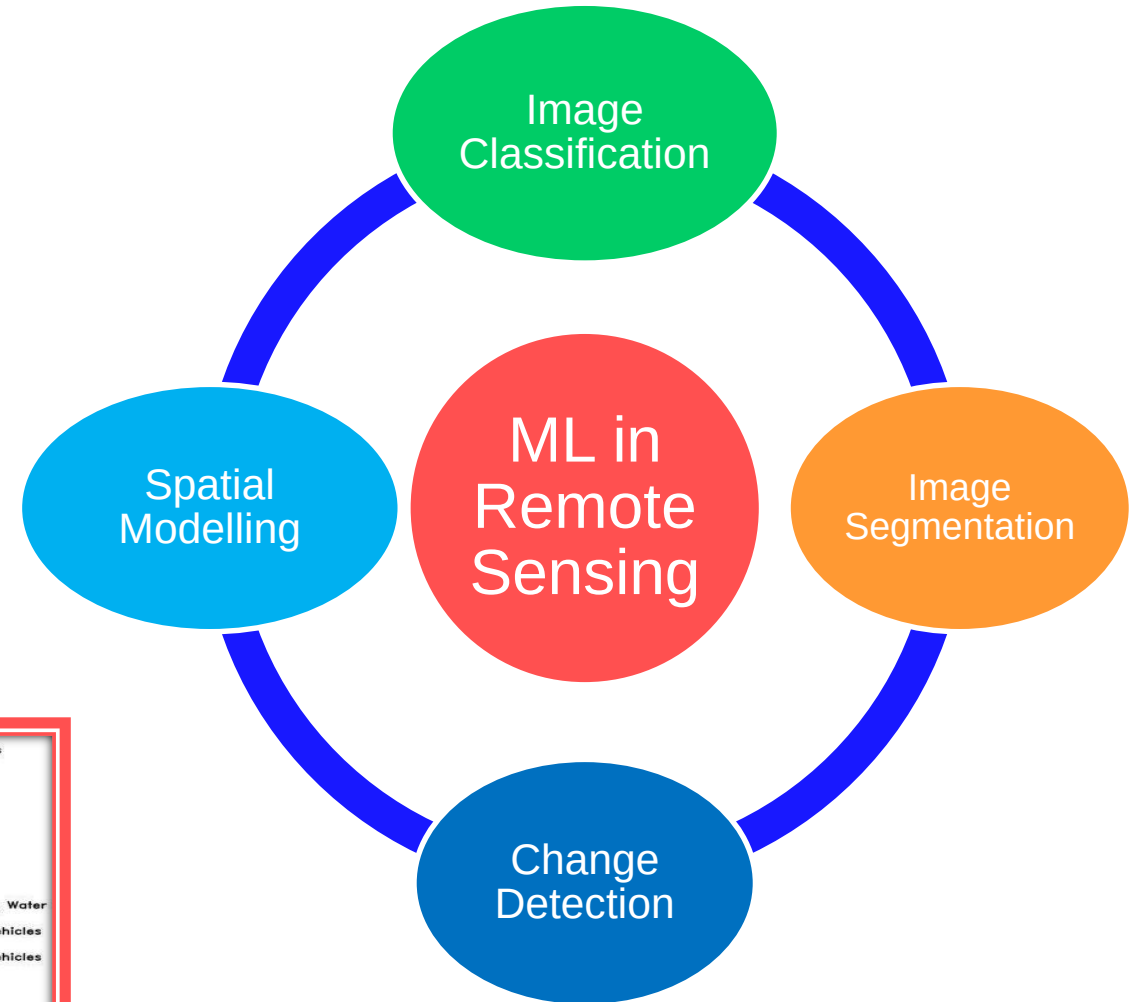
Artificial Intelligence Applications in GIS

Integration of AI with GIS technologies is enabling more efficient and accurate data processing, advanced spatial analysis, and automated informed decision-making.



Machine Learning (ML) Applications in GIS

ML algorithms have the capability to extract meaningful patterns and information from remote sensing data, automate analysis processes, and enhance the interpretation and utilization of geospatial data.



An example of image segmentation in satellite Imagery

Deep Learning (DL) Applications in GIS

Deep Learning (DL) is a subset of Machine Learning (ML) that focuses on training artificial neural networks with multiple layers.

Deep Learning models can automatically learn and make intelligent decisions by extracting hierarchical representations from large amounts of data.

Image Classification



Object Detection



Image Segmentation



Image Fusion



Deep learning use cases in Remote Sensing

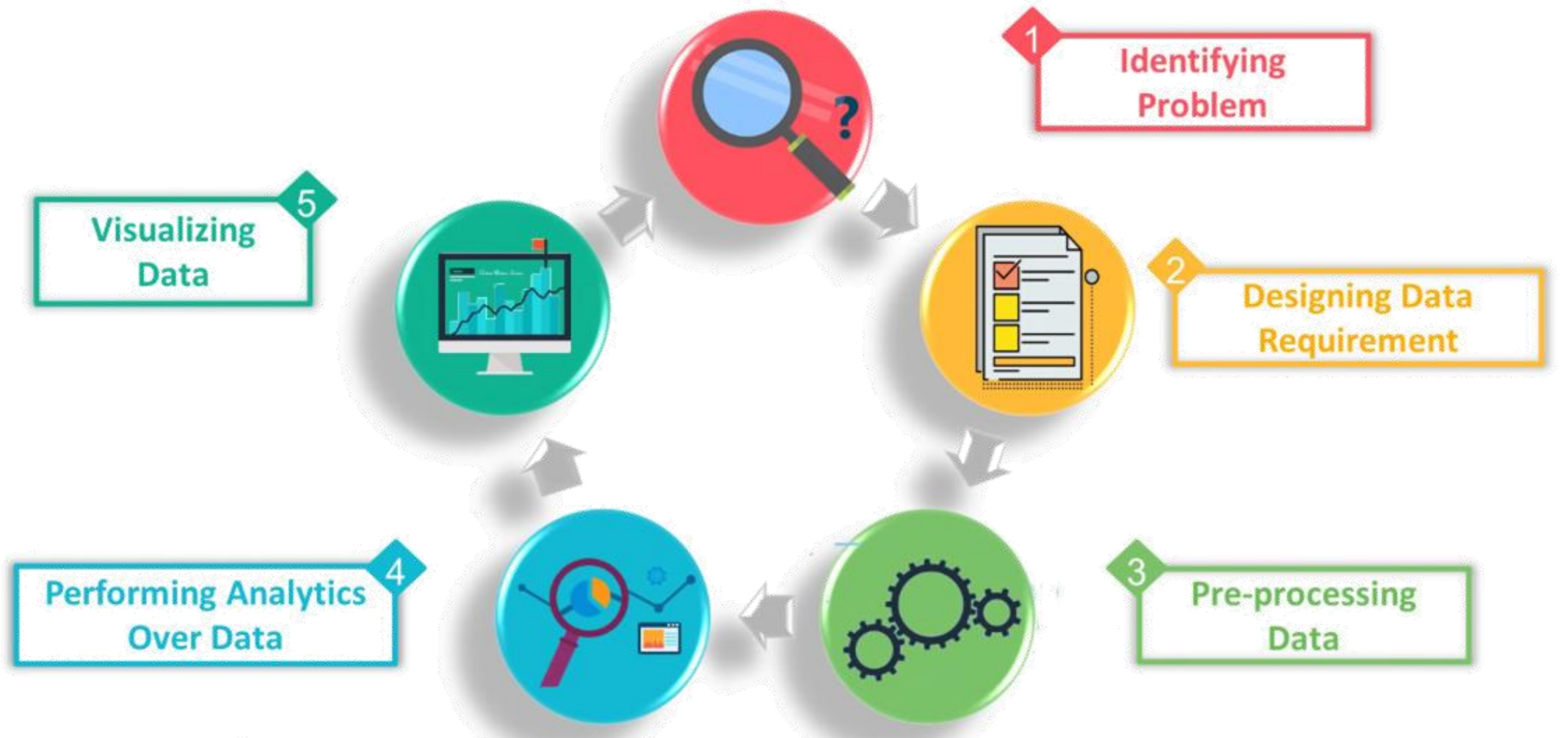
Big Data Analytics

Big data analytics examines large and different types of data (structured & unstructured) to uncover hidden patterns, correlations and other insights.

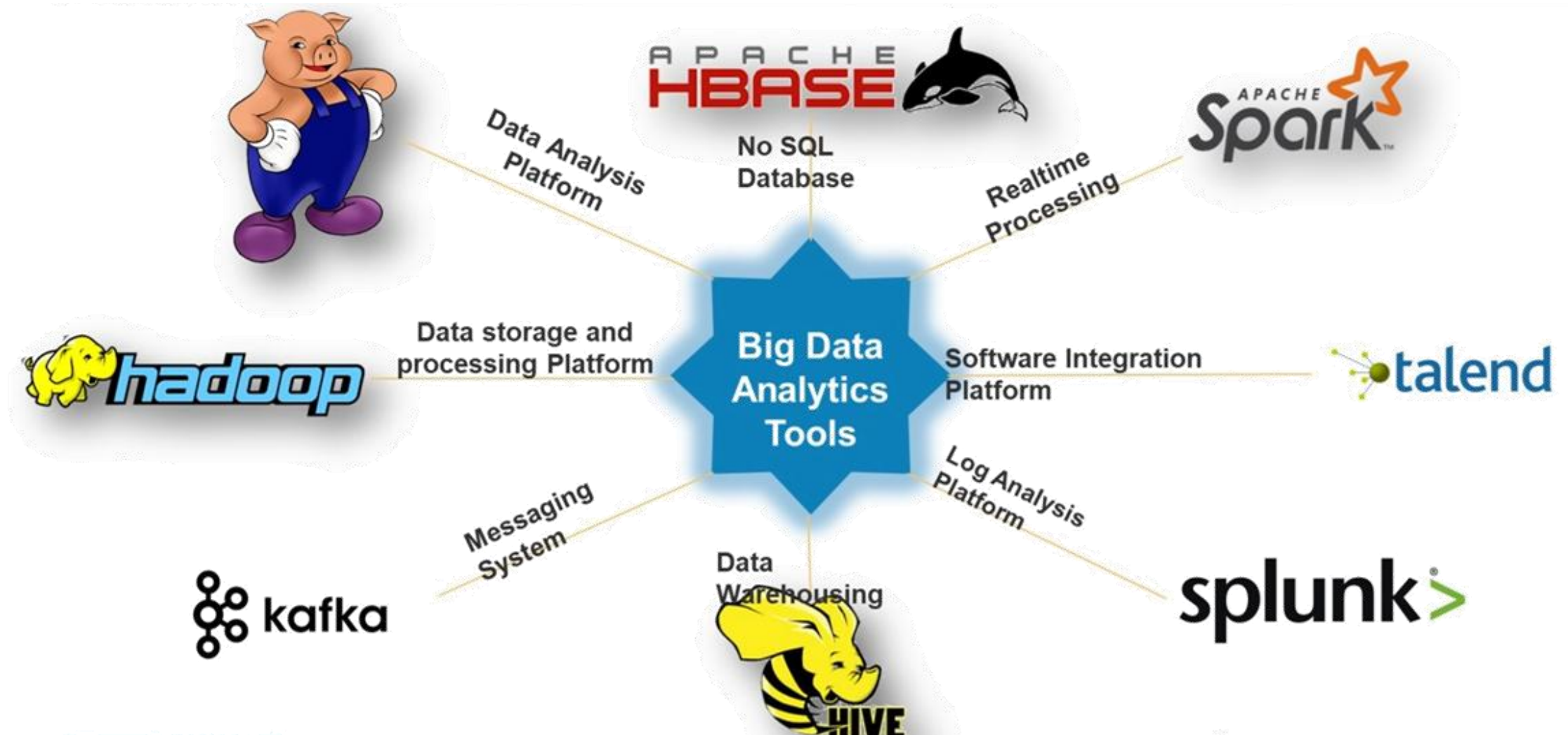
1. Descriptive analytics (What Happened and When)
2. Diagnostic analytics (Where and How it Happened)
3. Predictive analytics (What Will Happen and How)
4. Prescriptive analytics (What Should We Do)



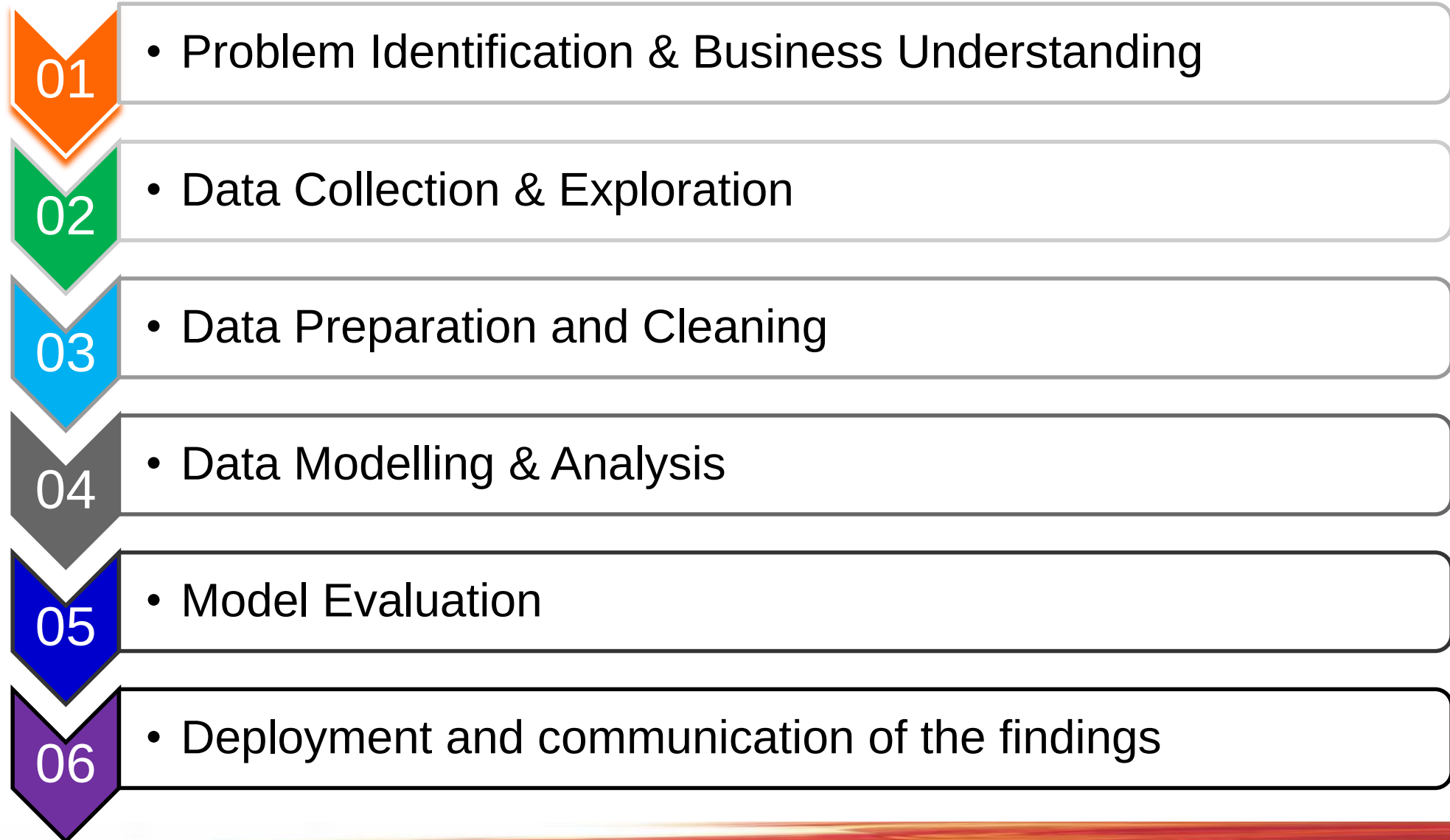
Stages in Big Data Analytics



Software tools used in Big Data Analytics



Data Science is Important



Thank you

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